



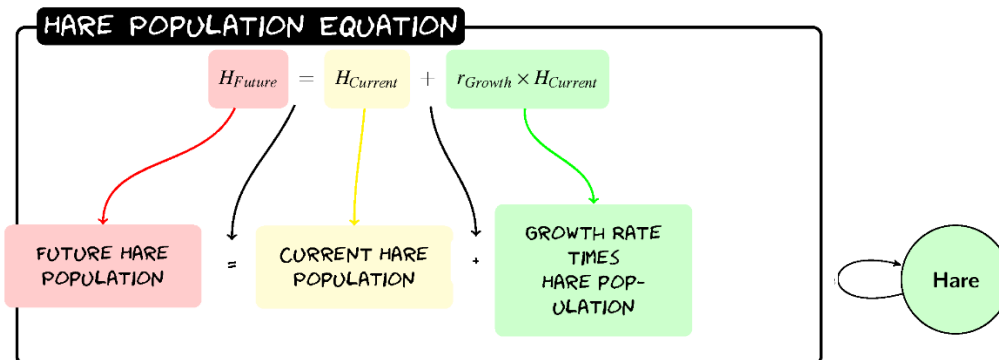
# The Maths of Predators and Preys

## Don't get Eaten by the Wolf

### Hares



No predators

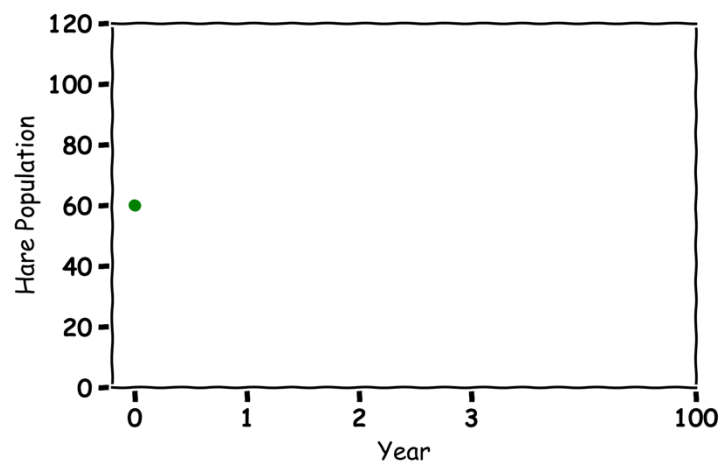


Given a population of 60 Hares and a **birth rate of 0.2** what would the population be after three years?

$$H_{Future} = H + 0.2(H)$$

| Year | Hare Population |
|------|-----------------|
| 0    | 60              |
| 1    |                 |
| 2    |                 |
| 3    |                 |

Sketch the Graph.

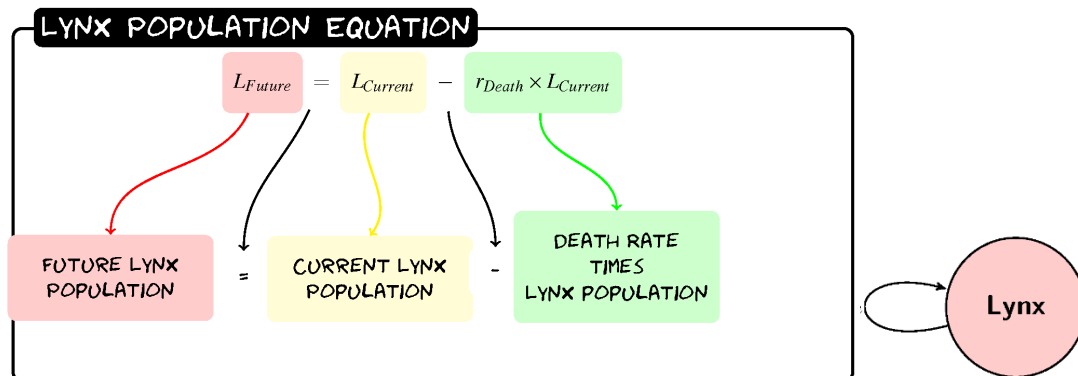


Guess what would happen in 100 years?



# Lynx

No Prey

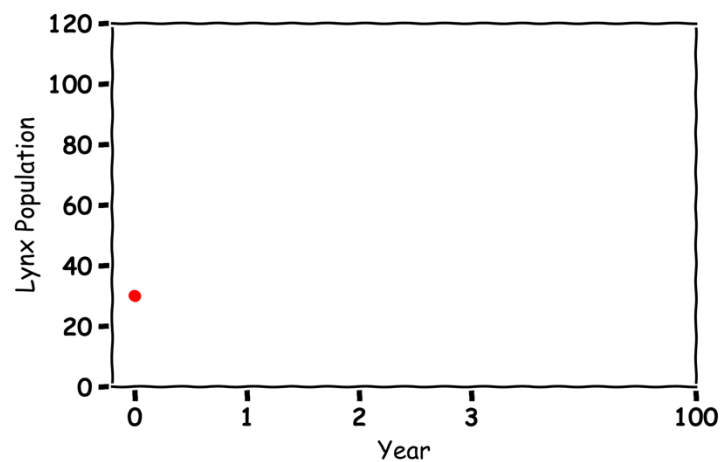


Given a population of 30 Lynxes and a **death rate of 0.1** what would the population be after three years?

$$L_{\text{Future}} = L - 0.1(L)$$

| Year | Lynx Population |
|------|-----------------|
| 0    | 30              |
| 1    |                 |
| 2    |                 |
| 3    |                 |

Sketch the Graph.

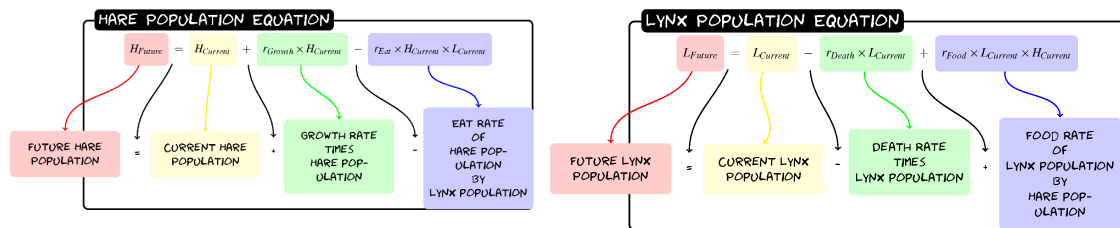
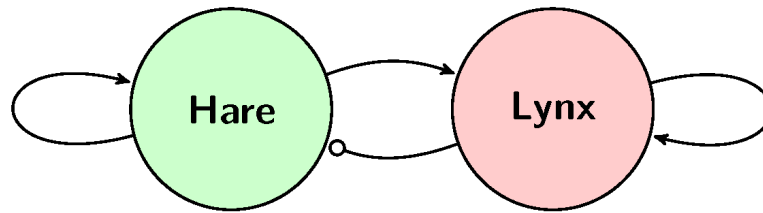


Guess what would happen in 100 years?



## Hares and Lynxes

Label the graph:



Given a population of 60 Hares and 30 Lynxes with a birth rate of 0.2, a death rate of 0.1, a eat rate of 0.005 and a food rate of 0.002 what would the populations be after three years?

| Hare  | Value | Lynx  | Value |
|-------|-------|-------|-------|
| Birth | 0.2   | Death | 0.1   |
| Eat   | 0.005 | Food  | 0.002 |

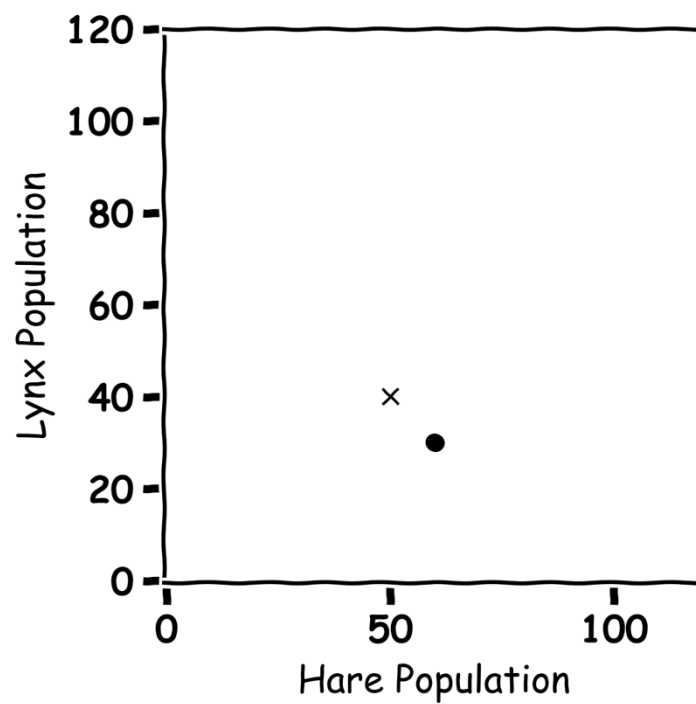
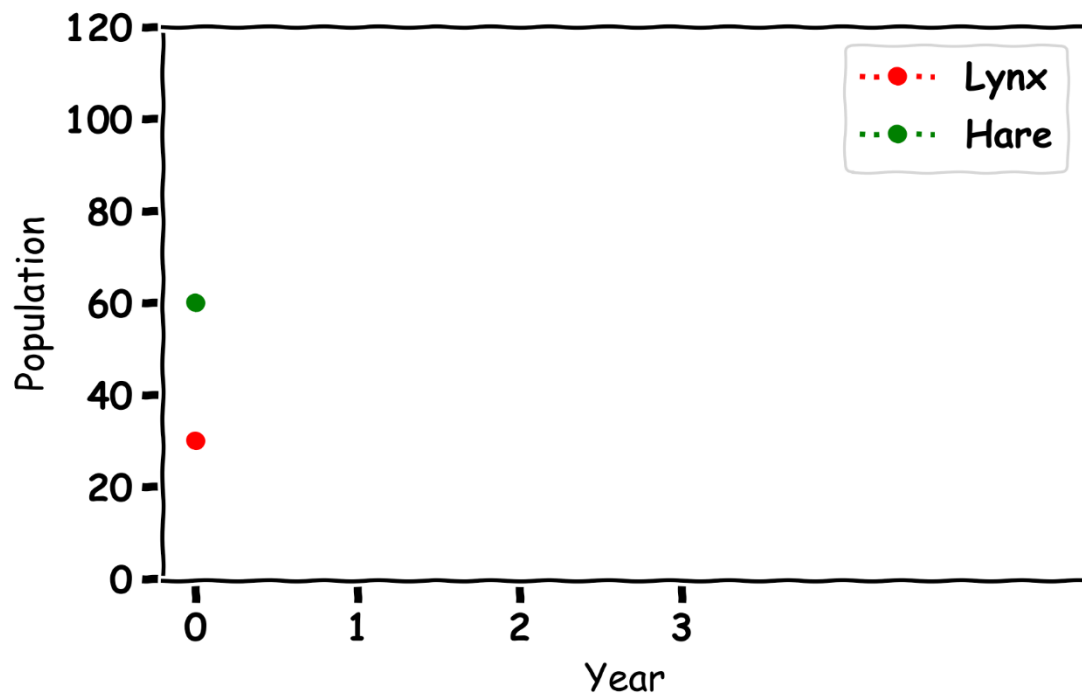
$$H_{\text{Future}} = H + 0.2(H) - 0.005(H)(L)$$

$$L_{\text{Future}} = L - 0.1(L) + 0.002(L)(H)$$

| Year | Hare Population | Lynx Population |
|------|-----------------|-----------------|
| 0    | 60              | 30              |
| 1    |                 |                 |
| 2    |                 |                 |
| 3    |                 |                 |



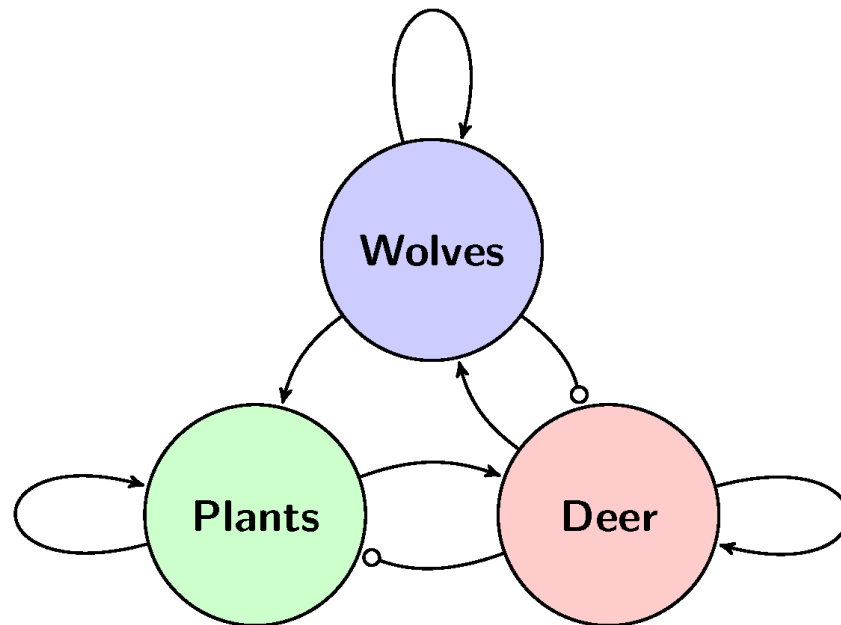
Sketch the graph:



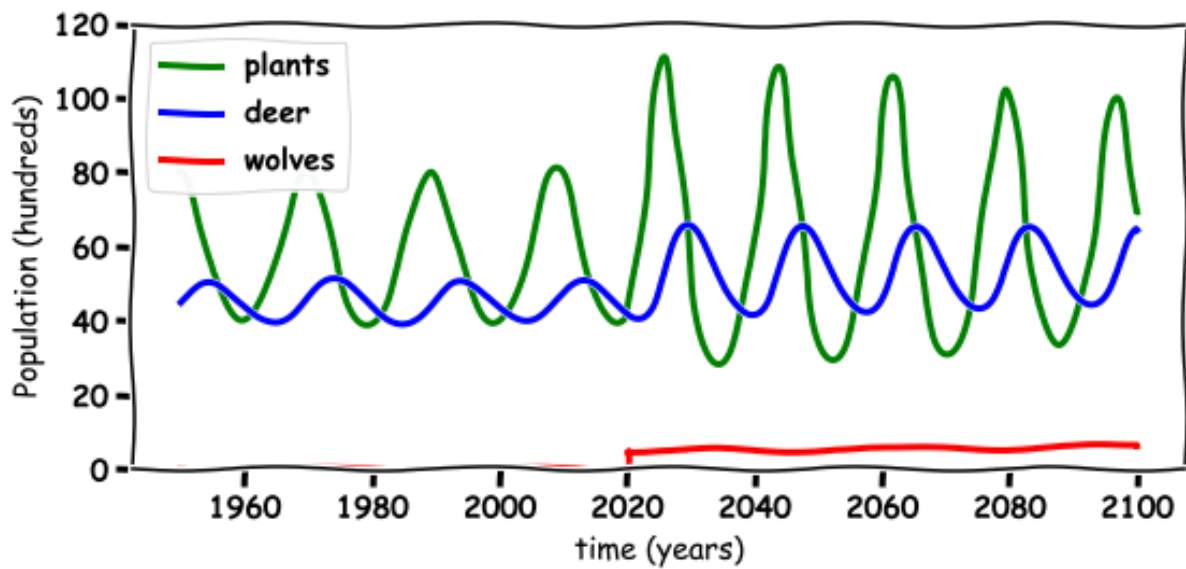


# Don't get Eaten by the Wolf

Label the graph



Describe what is happening in the below graph:





## Discussion Questions

1. What happened to the environment when the wolves are re-introduced?
2. What happened to the wolf population when they are re-introduced?
3. What happened to the deer population when the wolves are re-introduced?
4. Do you think the wolves should have been re-introduced in Ireland?
5. What else could you model with these Equations?